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NPS/003/009 - Technical Specification for 11kV, 20kV, 33kV, and 66kV Neutral Earthing Resistors

1. Purpose

This document is the technical specification for neutral earthing resistors (NERs) for use on the NEDL and YEDL distribution networks.

This document supersedes the following documents, all copies of which should be destroyed.

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NPS/003/009	1	Technical Specification for 11kV, 20kV, 33kV, and 66kV Neutral Earthing Resistors

2. Scope

This document covers the technical requirement for 11, 20, 33 and 66kV NERs for use with primary and grid transformers on the NEDL and YEDL distribution network, including both cable and busbar connected NERs, and also NERs fitted with isolating switches for use with Arc Suppression Coils (ASCs).

It will be necessary to consider and include any project specific requirements as detailed in Appendix 3, Addendum To Supplier Requirements.

3. Technical Requirements

3.1 Compliance with Specifications and Standards

Where reference is made within this specification to any other technical specification or standard, the current edition is intended.

3.2 General

NERs shall be designed, manufactured and tested to IEEE-32 and ENATS 41-36 as clarified or modified by technical specifications in Appendix 1 and the conditions listed below.

Equipment shall also comply with the Electromagnetic Compatibility (EMC) Regulations.

Appendix 2 contains a schedule of technical performance to be completed for all NER types.

Appendix 3, Addendum to Supplier Requirements, shall be completed with project specific requirements.

3.3 Enclosure

This shall be a free-standing, naturally air-cooled unit, suitable for outdoor use and not require maintenance for at least 30 years in a polluted/coastal environment according to EN ISO 12944-2 Category C4 including the heating effects of the resistor.

It shall have a minimum protection rating of IP33 to IEC 60529. The design shall be such that rainwater will freely run off and cannot collect at any point.

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All panels shall be securely fastened and not be capable of being removed without the use of tools.

Top mounted eyebolts shall be fitted for lifting the unit.

3.4 Rating Plate

A plate shall be securely fitted at eye level on the outside of the enclosure giving the following minimum information:

Manufacturers name and serial number

Date of manufacture

Nominal rated service voltage

Current rating

Time rating

Resistance value at 20°C with +/- tolerance

Enclosure IP rating

3.5 Earth Connections

A low voltage bushing and stud shall be provided through the enclosure wall for the resistor earth connection. A separate stud type terminal shall also be provided for the purpose of earthing the enclosure. These terminals shall be as close as practicable to ground level so as to minimise the length of earth conductor which will be exposed when the resistor is installed in position.

3.6 Enclosure Mounting

The NER shall be capable of being mounted on a concrete plinth.

3.7 Cable Boxes

All 11kV, 20kV and some 33kV units will be cable connected. For these units an air insulated cable box capable of accepting XLPE cables up to 500sq mm shall be provided, preferably on the longest side of the NER. The gland plate shall be removable, pre-drilled with an 88mm diameter hole and fitted with a suitable heat shrink cable gland. Where specified the box shall be capable of accommodating two ring core type current transformers (CT s). In this case the CT secondary wiring shall be terminated on a 6 way terminal block and an un-drilled gland plate shall be provided for outgoing multi-core cables.

Terminal blocks shall be either screw or stud type for use with an approved ring type crimped termination or spring loaded insertion type incorporating an approved hook type crimped connector. IPC or cage clamp designs will be accepted if they have been granted an ENA Notice of Conformity.

The box shall have a protection rating of IP54 to IEC 529 (BS EN60529).

3.8 HV Bushing

Some 33kV and all 66kV units will be connected by overhead busbar terminating onto an external bushing on the top of the unit. The bushing shall be in accordance with IEC60694.

Ground clearance of the bushing shall be as described in Appendix 1.

3.9 Resistor Element

Resistor elements shall be manufactured from a high temperature resistance alloy, preferably stainless steel, capable of withstanding wide temperature excursions whilst retaining mechanical strength. Ceramic insulators shall be used throughout.

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The temperature coefficient of resistance of the element material shall be such that the increase in resistance over the operating temperature range, when carrying rated current for the rated time, shall not exceed 18%.

The tolerance on the nominal resistance at 20°C shall not be greater than +/- 10%.

3.10 Isolating Switches (For use with Arc Suppression Coil systems)

Where the NER is required to operate in conjunction with an arc suppression coil the NER shall be equipped with an integral remotely controlled fault-make switch. The switch shall comply with ENATS 41-36 as supplemented by the technical specification in Appendix 1 of this document.

4. References

4.1 External Documentation

Reference	Title
ANSI/ IEEE Std 32-1972 (Reaffirmed 1984)	IEEE Standard Requirements, Terminology, and Test Procedure for Neutral Grounding Devices
ENATS 41-36	Distribution Switchgear For Service Up To 36kv (Cable And Overhead Conductor Connected)

4.2 Internal documentation

Reference	Title
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5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
K.Midgley	Standards Engineer	K.Midgley	19/04/07

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G.Bolden	Purchasing Manager	G.Bolden	30/04/07

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
D. Blackledge	Standards Engineer	D. Blackledge	25/04/07

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S.Johnson	DBD Administrator	S.Johnson	04/05/07

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C.Holdsworth	Standards and Assessment Manager	C.Holdsworth	30/04/07

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C.Morgan	Head of Procurement	C.Morgan	30/04/07

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Appendix 1: Technical Specification Sheet

NERs

	11kV				20kV		33kV		66kV		
	Type 1 B & C *	Type 2 B, BA, C & CA *	Type 3 B & 3 *	Type 4 B, BA, C, & CA *	Type 5 B & C *	Type 6 B, BA, C & CA *	Type 7 B & 7 *	Type 8 B & C *	Type 9 B *	Type10 B *	Type11 B *
	YEDL	YEDL	NEDL	NEDL	NEDL	NEDL	NEDL	NEDL	NEDL	NEDL	NEDL
Application	Urban	Rural									
Current Rating (A)	560	1500	600	1200	600	1200	600	1200	600	800	1000
Rated Voltage (kV)	11.5	11.5	11	11	20	20	33	33	66	66	66
Line Neutral Voltage (kV)	6.64	6.64	6.35	6.35	11.54	11.54	19.05	19.05	38.1	38.1	38.1
Rated Time (s)	10	10	17	13	17	13	10	10	10	10	10
Continuous Current (A)	0	0	0	0	0	0	0	0	0	0	0
Resistance @25°C (Ohm)	11.34	4.23	10.6	5.3	19.3	9.6	31.8	15.9	63.5	47.6	38.1
Max Element Temp Rise (°C)	1000 ⁰ C	1000 ⁰ C	1000 ⁰ C	1000 ⁰ C	1000 ⁰ C	1000 ⁰ C	1000 ⁰ C	1000 ⁰ C	1000 ⁰ C	1000 ⁰ C	1000 ⁰ C
Max increase in resistance at max temp.	+18%	+18%	+18%	+18%	+18%	+18%	+18%	+18%	+18%	+18%	+18%
Basic Impulse Level (IEC 60071) (kV)	75	75	75	75	110	110	150	150	250	250	250
Ground clearance to bushing top connector (m)	-	-	-	-	3.2	3.2	3.2	3.2	3.4	3.4	3.4
Ground Clearance to base of insulator (m)							2.45	2.45	2.45	2.45	2.45

* B = Busbar connected. C = Cable connected. A = For arc suppression coil.

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Switch to be incorporated in NERs for use with ASCs. (Types 2BA, 2CA, 4BA, 4CA, 6BA, 6CA)

	ENATS 41-36 Ref	Unit	11kV	20kV
Class	1.2		Outdoor	Outdoor
Rated Voltage (U_r)	1.4.1	kV	7.2	12
Frequency	1.4.3	Hz	50Hz	50Hz
No. of poles			1	1
Electrical endurance	3.1		E3	E3
Mechanical endurance	3.1		M2	M2
Lightning Impulse withstand (U_p)	1.4.2	kV	75	110
Short duration power frequency (U_d)	1.4.2	kV	28/32	38/45
Rated Normal Current	1.4.4	A	1500A for 10 sec 1200A for 13 sec	1200A for 10 sec
Rated short time withstand	1.4.5	kA	16	12.5
Rated duration of short circuit	1.4.7	Sec	3	3
Make duty	3.4.112	kA	16	12.5
Break duty	3.4.101	A	1500	1200
Arc extinguishing medium			SF6 or vacuum	SF6 or vacuum
Auxiliary and Control circuit Voltages	1.4.8 1.4.9	V	110v, 48v or 30v DC as specified	110v, 48v or 30v DC as specified
Gas monitoring (where appropriate)	1.5.9		Single stage switch	Single stage switch
Available Auxiliary Switches			5NO 5NC	5NO 5NC
Enclosure			Lockable	Lockable
Degree of protection	1.5.13		IP4X	IP4X
Position Indication			Mechanical or LED	Mechanical or LED

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APPENDIX 2: Technical Schedules to be completed by the manufacturer.

NERS

	11kV				20kV		33kV		66kV		
	Type 1	Type2	Type 3	Type 4	Type 5	Type 6	Type 7	Type 8	Type 9	Type 10	Type 11
	YEDL	YEDL	NEDL	NEDL	NEDL	NEDL	NEDL	NEDL	NEDL	NEDL	NEDL
Current Rating	560A	1500A	600A	1200A	600A	1200A	600A	1200A	600A	800A	1000A
HV Connection type (cable, busbar or both)											
Rated voltage (V)											
Rated time (s)											
Resistance and tolerance @ 25°C											
Continuous current (A)											
Max temperature rise @ rated time and current											
BIL											
Element type											
Element material											
Temperature coefficient of resistance (% per deg C)											
Resistance change for 1000 °C rise (%)											

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	11kV				20kV		33kV		66kV		
	Type 1	Type2	Type 3	Type 4	Type 5	Type 6	Type 7	Type 8	Type 9	Type 10	Type 11
	YEDL	YEDL	NEDL	NEDL	NEDL	NEDL	NEDL	NEDL	NEDL	NEDL	NEDL
Current Rating	560A	1500A	600A	1200A	600A	1200A	600A	1200A	600A	800A	1000A
Total inductance (mH)											
HV bushing Voltage class (kV) BIL (kV)											
LV bushing Voltage class (kV) BIL (kV)											
NER housing material Thickness (mm) Finish											
Ingress protection (IEC 529) NER Cable Boxes											
Cable Gland Type (where appropriate)											
HV Bushing connection ground clearance (m)											
HV Bushing base ground clearance (m)											
Overall dimensions (mm) Width Depth Height											

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	11kV				20kV		33kV		66kV		
	Type 1	Type2	Type 3	Type 4	Type 5	Type 6	Type 7	Type 8	Type 9	Type 10	Type 11
	YEDL	YEDL	NEDL	NEDL	NEDL	NEDL	NEDL	NEDL	NEDL	NEDL	NEDL
Current Rating	560A	1500A	600A	1200A	600A	1200A	600A	1200A	600A	800A	1000A
Weight (kg)											
Drawing No											
Compliance with EMC regulations (Yes/No)											

Switch to be incorporated in NERs to be installed as part of an Arc Suppression Coil Scheme

	ENATS 41-36 Ref	Unit	11kV	20kV
Class	1.2	Indoor/Outdoor		
Rated Voltage (U_r)	1.4.1	kV		
Frequency	1.4.3	Hz		
No. of poles				
Class	3.1			
Lightning Impulse withstand (U_p)	1.4.2	kV		
Short duration power frequency impulse withstand (U_d)	1.4.2	kV		
Rated Normal Current	1.4.4	A		
Rated short time withstand	1.4.5	kA		
Rated duration of short circuit	1.4.7			
Make duty	3.4.112	kA		
Break duty	3.4.101	A		
Arc extinguishing medium				
Auxiliary and Control circuit Voltages	1.4.8 1.4.9			
Drive mechanism	1.5.5			
Gas monitoring (where appropriate)	1.5.9			

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Available Auxiliary Switches				
Enclosure				
Degree of protection	1.5.13			
Position indication				

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Appendix 3: Addendum To Supplier Requirements

Project specific requirements will be provided by Primary Engineering Projects for inclusion in this appendix.